

Identification of Systematic Monetary Policy

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Introduction

- This paper proposes an **identification design** to estimate the effects of **systematic monetary policy** on the propagation of macroeconomic shocks.
- **Systematic monetary policy** is the purposeful response by policymakers to macroeconomic shocks.
- We measure it here by the composition of **hawks** and **doves** in the FOMC.
- Members of the FOMC perceived to be more concerned with:
 - **inflation** are called **hawks**.
 - **employment and growth** are called **doves**.
- The classification of FOMC members as hawks or doves is based on narratives from news archives, portraying them as either hawks or doves, as in **Istrefi (2019)**.

Introduction

- There is the concern that the fluctuations in the composition of hawks and doves in the FOMC are endogenous to economic and political developments.
⇒ Construct an **instrument** that exploits the mechanical rotation of voting rights in the FOMC.
- The identification design is then applied to address a classical question in macroeconomics:

How do the effects of fiscal policy depend on the response of monetary policy?

- Main results:

	FFR	GDP	Fiscal multiplier
Dovish FOMC	Delayed tightening	Expansion	2 and 3
Hawkish FOMC	Faster tightening	Contraction	Below 0

Outline

1. Stylized example to illustrate endogeneity problems.
2. Identification design:
 - Measure of systematic monetary policy: The Hawk-Dove balance.
 - FOMC rotation instrument: Relevance and Exogeneity.
 - Local projection framework.
3. Estimates of the causal effects of systematic monetary policy on:
 - Real GDP and real government spending in response to fiscal spending shocks.
 - Fiscal multiplier.
 - Interest rate response.

Stylized example: Identification challenge in a New Keynesian Model

What if sys. monetary policy fluctuates endogenously?

Textbook New Keynesian model, except for a monetary policy rule with time-varying coefficients. We show that if systematic monetary policy fluctuates endogenously, then the estimates on the effects of fiscal spending shocks are biased.

Households	$\max_{\{C_t, N_t, B_t\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t \left(\log C_t - N_t^{1+\varphi} \right) \text{ s.t. } BC$
Intermediate firms	$Y_{it} = x_t^a N_{it}$
Final goods firms	$Y_t = \left(\int_0^1 Y_{it}^{(\epsilon-1)/\epsilon} di \right)^{\epsilon/(\epsilon-1)}$
Fiscal policy	$G_t = \gamma Y x_t^s$
Shocks	$\log x_t^a = \rho_a \log x_{t-1}^a + \varepsilon_t^a, \text{ with } \varepsilon_t^a \sim (0, \sigma_a^2)$ $\log x_t^s = \rho_s \log x_{t-1}^s + \varepsilon_t^s, \text{ with } \varepsilon_t^s \sim (0, \sigma_s^2)$
Monetary policy	$\dot{i}_t = \tilde{\phi}_t \pi_t, \text{ where } \tilde{\phi}_t \text{ is systematic monetary policy.}$ $\phi_t = \rho_\phi \phi_{t-1} + \zeta^s \varepsilon_t^s + \zeta^a \varepsilon_t^a + \eta_t, \text{ where } \phi_t = (\tilde{\phi}_t - \phi).$

Monetary Policy rule

- η_t captures **exogenous** changes in systematic monetary policy, and we allow systematic monetary policy to be **endogenous** since ϕ_t may respond to macroeconomic shocks $(\varepsilon_t^s, \varepsilon_t^a)$.
- Using a first order approximation, the equilibrium dynamics of GDP are given by:

$$y_t = a + b_s x_t^s + b_a x_t^a + c_s x_t^s \phi_t + c_a x_t^a \phi_t + d \phi_t,$$

where a, b_s, b_a, c_s, c_a, d depend on the structural parameters.

- Suppose we want to identify the effects of monetary policy on the response to a fiscal spending shock from a regression. We observe $\{y_t, \varepsilon_t^s, \phi_t\}$, and estimate the state-dependent local projection

$$y_{t+h} = \alpha^h + \beta^h \varepsilon_t^s + \gamma^h \varepsilon_t^s \phi_t + \delta^h \phi_t + v_{t+h}^h$$

for $h = 0, \dots, H$ forecast horizons.

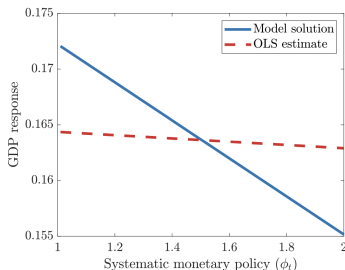
Inconsistency of OLS estimates

- v_{t+h}^h contains lagged spending shocks, contemporaneous and lagged technology shocks, the interactions of these shocks with ϕ_t , and for $h > 0$ also $(\varepsilon_t^s, \varepsilon_t^a)$ and policy shifter η_t between t and $t + h$.
- Consistency of $(\hat{\beta}^h, \hat{\gamma}^h, \hat{\delta}^h)$ only holds under the **strong exogeneity assumption**

$$\zeta^a = \zeta^s = 0.$$

that is, if ϕ_t is independent of the macroeconomic shocks.

- Example with ϕ_t correlated with both shocks ($\zeta^s = 1, \zeta^a = 0.25$):



- OLS estimate: $\hat{\beta}^0 + \hat{\gamma}^0 \phi_t$
- Model solution: $b_s + c_s \phi_t$

Consistency of IV

- Suppose we observe an estimate ϕ_t^{IV} that is:
 - **Relevant:** Correlated with ϕ_t .
 - **Exogenous:** Uncorrelated with all past, present and future $(\varepsilon_t^a, \varepsilon_t^s)$ and uncorrelated with all past and future values of η_t .
- When we use $(\varepsilon_t^s, \varepsilon_t^s \phi_t^{IV}, \phi_t^{IV})$ as the instrument vector for the regressors $(\varepsilon_t^s, \varepsilon_t^s \phi_t, \phi_t)$, we get consistent estimates for $(\beta^h, \gamma^h, \delta^h)$, even when ϕ_t fluctuates endogenously in response to macroeconomic shocks ($\zeta^a, \zeta^s \neq 0$).
- This result guides the rest of the paper, in which an instrument for systematic monetary policy is proposed and used to estimate the causal effects of systematic monetary policy.

Identification Design

Identification Design

- We want to study how systematic monetary policy in the U.S. shapes the propagation of macroeconomic shocks.
- The identification design relies on three elements:
 1. A **measure** of systematic monetary policy: The Hawk-Dove balance in the FOMC.
 2. An **instrument** for systematic monetary policy.
 3. A state-dependent **local projection** that combines 1 and 2.

Hawk-Dove balance in the FOMC: Constructing the measure

- The FOMC
 - is the committee of the Fed that sets U.S. monetary policy,
 - consists of 12 members: all 7 members of the Board of Governors of the Federal Reserve System, including the Fed Chair, the president of the FRB of New York, and 4 of the remaining 11 FRB presidents, who serve one-year terms on a rotating basis.
- Individual policy preferences.
 - Policy preferences of individual FOMC members are measured using the [Istrefi \(2019\)](#) classification of members as **hawks** or **doves**, for the period 1960-2023.
- Aggregate Hawk-Dove balance.
 - To measure variation in systematic monetary policy over time, the cross-section of individual FOMC member preferences are aggregated into a Hawk-Dove balance for each meeting.

Individual policy preferences

- We define $Hawk_{i\tau}$ as the policy preference of FOMC member i at meeting τ :

$$Hawk_{i\tau} = \begin{cases} +1 & \text{Consistent hawk} \\ +\frac{1}{2} & \text{Swinging hawk} \\ 0 & \text{Preference unknown} \\ -\frac{1}{2} & \text{Swinging dove} \\ -1 & \text{Consistent dove} \end{cases}$$

- Istrefi (2019) uses more than 20,000 real-time media articles and business reports of Fed watchers to categorize individual FOMC members as hawks or doves for each FOMC meeting based on the news information available up until the meeting. [▶ More](#)
- Through the lens of the stylized model, **hawks would prefer a larger ϕ_t than doves.**

Aggregate Hawk-Dove balance

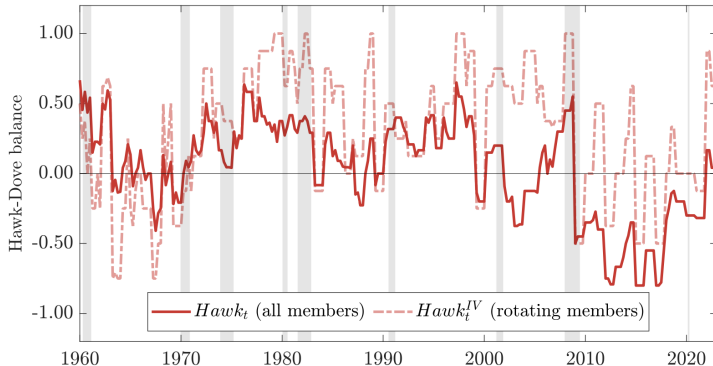
- The aggregate Hawk-Dove balance is computed by

$$Hawk_{\tau} = \frac{1}{|\mathcal{M}_{\tau}|} \sum_{i \in \mathcal{M}_{\tau}} Hawk_{i\tau}$$

where $\mathcal{M}_{\tau}$ is the set of FOMC members at meeting τ .

- $Hawk_{\tau}$ is then aggregated from meeting to quarterly frequency by using the average balance in the first month of the quarter.
- Measuring systematic policy via $Hawk_t$ has several advantages:
 - No need to specify a particular reaction function, or to restrict the analysis to specific policy instruments.
 - Independently of the policy tool or policy rule, this measure reflects the aggressiveness of the FOMC towards fulfilling one or the other leg of the dual mandate.
 - $Hawk_t$ reflects public beliefs, in real-time, about monetary policymakers, so it doesn't use ex-post information not available at the time of the policy decisions.

Evolution of the Hawk-Dove balance 1960-2023



- Considerable variation, featuring both hawkish and dovish majorities. Matches well with narratives of U.S. monetary policy.
- The variation reflects the turnover of rotating FOMC members, the turnover of non-rotating FOMC members, and changes in policy preferences of incumbent FOMC members.

Potential Endogeneity of $Hawk_t$

- Systematic monetary policy may change depending on the **state of the economy**. For example, the Fed may become more dovish in response to high unemployment, or more hawkish in response to high inflation.
- Changes in systematic monetary policy may also be driven by **political pressure**, through swings of incumbent FOMC members and through new appointments.
- To address the endogeneity of the Hawk-Dove balance, this paper proposes an instrument which leverages **exogenous variation** in $Hawk_t$ that arises from the annual FOMC rotation.

FOMC Rotation Instrument

- Each year, 4 FOMC memberships rotate among 11 FRB presidents following a mechanical scheme in place since the 1940's.
- According to the scheme, some FRB presidents become FOMC members every second year (Cleveland and Chicago) and others every third year (Philadelphia, Richmond, Boston, Dallas, Atlanta, St.Louis, Minneapolis, San Francisco and Kansas City).
- As the rotation of voting rights is independent of the state of the economy, it **induces exogenous variation** in $Hawk_t$.

FOMC Rotation Instrument

- The instrument is given by:

$$Hawk_{\tau}^{IV} = \frac{1}{|\mathcal{R}_{\tau}|} \sum_{i \in \mathcal{R}_{\tau}} Hawk_{i\tau}$$

where $\mathcal{R}_{\tau}$ denotes the set of rotating FOMC members at FOMC meeting τ .

- A full set of rotating members consists of $|\mathcal{R}_{\tau}| = 4$ members.
- The FOMC rotation instrument is then aggregated to quarterly frequency analogously to the Hawk-Dove balance.
- We now argue that the instrument satisfies the **relevance** and **exogeneity** assumptions.

Relevance of instrument

- $Hawk_t^{IV}$ aggregates the policy preferences of one-third of the FOMC members, capturing a significant part of the variation in the overall Hawk-Dove Balance. In fact, $Corr(Hawk_t, Hawk_t^{IV}) = 0.64$.
- Applying the weak instrument test from Montiel Olea and Pflueger (2013), the null hypothesis of a relative weak instrument bias exceeding 5% is rejected.
- Extensive margin changes in the FOMC composition due to the rotation account for 53% of the variance in yearly changes of $Hawk_t$.

Exogeneity of instrument

- The rotation scheme is mechanical and time-invariant, and therefore unrelated to the state of the economy.
- New appointments of FRB presidents are relatively infrequent and unlikely to be influenced by the federal government.
- Swings of FRB presidents between being a hawk or dove could be driven by macroeconomic shocks. However, swings occur only in 1.3% of member-meetings pairs, and they account for only a negligible fraction of the variance.
- $Hawk_t^{IV}$ displays relatively short-lived time series fluctuations that are unlikely to be correlated with slow-moving macroeconomic trends, and it is uncorrelated with business cycle fluctuations.

Local Projection Framework

Local Projection Framework

- The local projection framework with the rotation instrument permits **causal identification** of how systematic monetary policy shapes the propagation of various macroeconomic shocks.
- We regress an outcome variable of interest, x_{t+h} , on a macroeconomic shock of interest, ε_t^S , the interaction of the shock with $Hawk_t$ as well as $Hawk_t$ in levels and a vector of additional control variables Z_{t-1} :

$$x_{t+h} = \alpha^h + \beta^h \varepsilon_t^S + \gamma^h \varepsilon_t^S (Hawk_t - \overline{Hawk}) \\ + \delta^h (Hawk_t - \overline{Hawk}) + \zeta^h Z_{t-1} + v_{t+h}$$

for $h = 0, \dots, H$ forecast horizons.

- To address the potential endogeneity of $Hawk_t$, we use the instrument vector

$$q_t = [1, \varepsilon_t^S, \varepsilon_t^S (Hawk_t^{IV} - \overline{Hawk}^{IV}), (Hawk_t^{IV} - \overline{Hawk}^{IV}), Z_{t-1}]$$

Local Projection Framework

$$x_{t+h} = \alpha^h + \beta^h \varepsilon_t^s + \gamma^h \varepsilon_t^s (\text{Hawk}_t - \overline{\text{Hawk}}) \\ + \delta^h (\text{Hawk}_t - \overline{\text{Hawk}}) + \zeta^h Z_{t-1} + v_{t+h}$$

- The two key coefficients are β^h and γ^h .
 - β^h captures the average response, when the Hawk-Dove balance equals its sample average.
 - γ^h captures the differential response, when the FOMC is more or less hawkish than the sample average.
- This framework can be used to study the propagation of any shock through systematic U.S. monetary policy.

Validation exercise for $Hawk_t$ and $Hawk_t^{IV}$

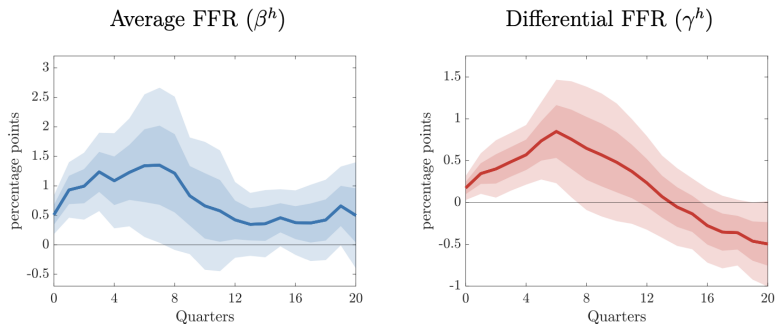
- Is $Hawk_t$ really measuring systematic monetary policy? Given our definition of hawkish policy makers, we should expect a **hawkish** FOMC to respond more **aggressively** to inflation.
- As a validation exercise, we empirically test this correlation via a dynamic Taylor rule regression:

$$FFR_{t+h} = \alpha^h + \beta^h \hat{\pi}_t + \gamma^h \hat{\pi}_t (Hawk_t - \overline{Hawk}) + \zeta^h Z_{t-1} + v_{t+h}^h$$

for $h = 0, 1, \dots, H$, FFR_{t+h} denotes the federal funds rate and $\hat{\pi}_t$ the average of the one- and two-quarter ahead Greenbook inflation forecast.

- The control vector includes four lags of the federal funds rate and the inflation forecast.

Validation exercise for $Hawk_t$ and $Hawk_t^{IV}$



Notes: The figure shows responses of the federal funds rate to an inflation Greenbook forecast that is one percentage point above its sample average, conditional on systematic monetary policy ($Hawk_t$). We show IV estimates based on (E.1). The β^h captures the responses when $Hawk_t$ equals its sample average. The γ^h captures the differential responses when $Hawk_t$ exceeds the sample average by two hawks. The shaded areas indicate 68% and 95% confidence bands using Newey-West standard errors.

Figure: FFR response to inflation and FOMC hawkishness

A hawkish FOMC raises the FFR significantly more aggressively in the presence of higher inflation forecasts.

Government Spending and Monetary Policy

Data for government spending shocks

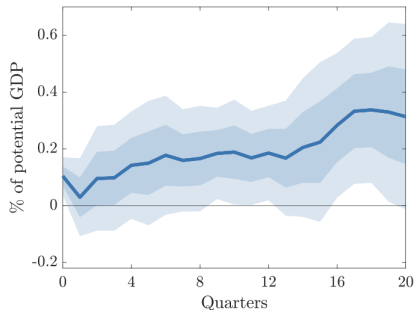
- Now we use the identification design to estimate how the effects of U.S. government spending shocks depend on systematic monetary policy.
- The baseline shock of interest, ε_t^S , is the military spending shock constructed by [Ramey \(2011\)](#) and [Ramey and Zubairy \(2018\)](#).
- The shock is constructed as the present value of expected changes in real defense spending over the next years, typically up to a horizon of five years, and expressed relative to real potential GDP.
- The two variables of interest, x_{t+h} are real GDP and real government spending, both expressed relative to real potential GDP.
- Sample covers the period from 1960Q1 to 2014Q4.

Identifying assumptions

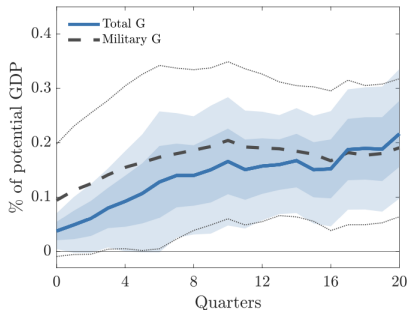
- Two key assumptions are necessary for the causal interpretation of the estimates of β^h and γ^h .
 1. The FOMC rotation instrument is orthogonal to all macroeconomic shocks at all leads and lags.
 2. Military spending shocks are random shocks. In particular, the distribution of military spending shocks may not depend on systematic monetary policy.
- According to Ramey and Shapiro (1998) and Ramey and Zubairy (2018), military spending shocks are unanticipated changes in spending plans triggered by geopolitical events and are therefore exogenous to the economy.
- One concern is that military spending shocks might be timed to episodes with a more dovish FOMC. We find no significant effects of the Hawk-Dove balance on contemporaneous or future military spending shocks.

IV estimates: When $Hawk_t$ equals its sample average

(a) Average GDP (β^h)



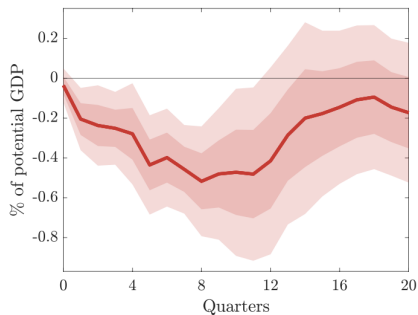
(b) Average G (β^h)



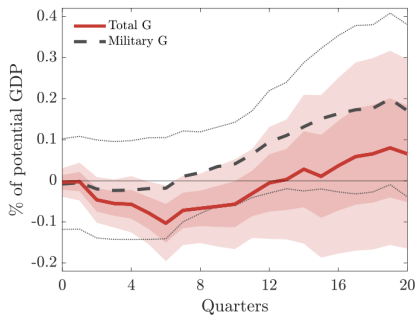
- The average responses of GDP and G are positive and significantly different from zero at most horizons beyond the first year.
- Both responses build up gradually and exceed 0.15% for GDP and 0.11% for total G after one year.

IV estimates: Differential responses when the FOMC exceeds the average Hawk-Dove balance by two hawks

(c) Differential GDP (γ^h)



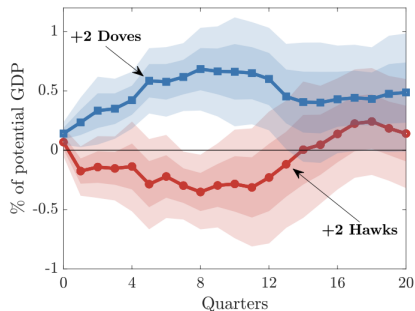
(d) Differential G (γ^h)



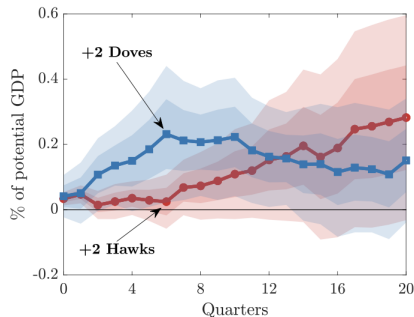
- The GDP response is lower after a fiscal expansion when the FOMC is more hawkish. This effect is statistically significant at the 5% level until three years after the shock.

IV estimates: State-dependent responses when $Hawk_t$ exceeds the sample average by two hawks or doves

(e) State-dependent GDP ($\beta^h \pm \gamma^h$)



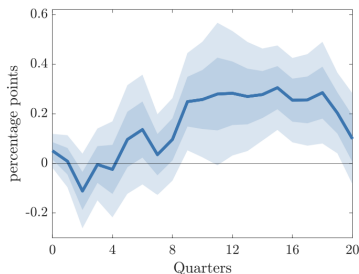
(f) State-dependent G ($\beta^h \pm \gamma^h$)



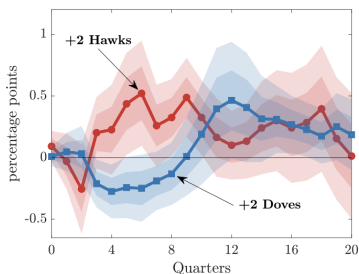
- The dovish FOMC supports the GDP expansion while the hawkish FOMC undoes the GDP expansion.

Inspecting the mechanism: FFR response

(a) Average FFR (β^h)



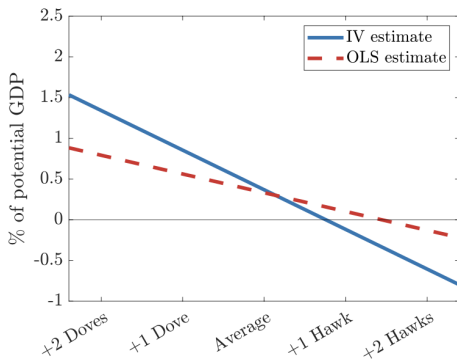
(b) State-dependent FFR ($\beta^h \pm \gamma^h$)



- FFR increases faster and more strongly under a hawkish FOMC.
- The delayed FFR response is consistent with the initial uncertainty surrounding the military spending shock.

Ignoring endogeneity of $Hawk_t \implies$ Biased conclusions

- Overall, the evidence suggests that monetary offset of fiscal spending shocks is not a constant feature of monetary policy but varies strongly with the Hawk-Dove balance in the FOMC.
- Comparison with OLS:



The displayed estimates are: $\sum_{h=1}^4 [\beta^h + \gamma^h (Hawk_t - \overline{Hawk})]$.

Fiscal spending multiplier

- A key object for the design and evaluation of fiscal policies is the **fiscal spending multiplier**.
- Using the same framework, we can estimate how it depends on the hawkishness of the FOMC.
- The multiplier is defined as the dollar amount by which GDP increases per dollar increase in fiscal spending (both in real terms).
- To study how systematic monetary policy shapes the fiscal multiplier, we define the **monetary policy-dependent** multiplier:

$$FM^H(\chi) = \frac{\sum_{h=0}^H (\beta_{GDP}^h + \gamma_{GDP}^h \chi)}{\sum_{h=0}^H (\beta_G^h + \gamma_G^h \chi)}$$

where χ indicates some value of the Hawk-Dove balance in deviation from the average.

Fiscal spending multiplier: Results

	Baseline model					Linear
Outcome	+2 Hawks	+1 Hawk	Average	+1 Dove	+2 Doves	model
<i>Two-year horizon</i>						
Multiplier	-4.825 (5.229)	-0.476 (1.418)	1.348 (0.708)	2.351 (0.934)	2.986 (1.239)	0.860 (1.427)
GDP (<i>cum</i>)	-1.689 (0.989)	-0.282 (0.768)	1.124 (0.649)	2.531 (0.689)	3.937 (0.865)	0.616 (1.057)
G (<i>cum</i>)	0.350 (0.250)	0.592 (0.300)	0.834 (0.395)	1.076 (0.510)	1.319 (0.634)	0.716 (0.338)
<i>Four-year horizon</i>						
Multiplier	-1.790 (2.637)	-0.001 (0.862)	1.308 (0.475)	2.307 (0.808)	3.095 (1.162)	0.838 (1.449)
GDP (<i>cum</i>)	-2.735 (2.498)	-0.002 (1.557)	2.731 (0.842)	5.465 (1.045)	8.198 (1.892)	1.494 (2.747)
G (<i>cum</i>)	1.528 (1.010)	1.808 (0.804)	2.088 (0.734)	2.368 (0.848)	2.649 (1.079)	1.782 (0.689)

Conclusion

- This paper proposes an identification design to estimate the effects of systematic monetary policy on the propagation of macroeconomic shocks.
- This design consists of:
 - Constructing a measure of systematic monetary policy using the narrative classification of FOMC members' policy preferences from [Istrefi \(2019\)](#).
 - A novel FOMC rotation instrument for systematic monetary policy.
- Many research opportunities, such as revisiting the effects of various fiscal, technology, and oil shocks and their dependence on systematic monetary policy.
- Using this identification design, we study government spending shocks in the U.S. and find that fiscal spending multipliers depend strongly and significantly on systematic monetary policy.

Thank you!

Appendix

Appendix - Individual policy preferences in more detail

- 129 of the 147 FOMC members between 1960-2023 are classified as hawk or dove.
- The news coverage for the remaining 18 members does not allow classification for any meeting.
- The majority (95) of the classified FOMC members are consistently hawks or doves over time. The rest switches camps at least once.
- Swings are equally split in either direction and quite uniformly distributed over time.
- On average, the 34 swinging FOMC members switch camps at only 1.8% of the member-meeting pairs.
- While true preferences are unobserved, [Istrefi \(2019\)](#) shows that perceived preferences match well with policy tendencies unknown in real-time to the public, expressed by preferred interest rates, forecasting patterns of individual FOMC members, and dissents.